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Subject Code

PT I - INTERDISCIPLINARY PROJECT

SMART COMPLAINT & GRIEVANCE REDRESSAL PORTAL

**(Hierarchical Campus Grievance Redressal &
SLA Auto-Escalation Web Platform)**

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Internal Guide

Ms. R. Shanumgaa Priya, M.E.



PRESENTATION OUTLINE

- COURSE CERTIFICATE
- ABSTRACT
- OBJECTIVES
- SYSTEM ARCHITECTURE
- PROJECT IMPLEMENTATION
- RESULTS AND DISCUSSIONS
- RESULT SNAPSHOTS
- CONCLUSION & FUTURE WORK
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COURSE CERTIFICATE

SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE)

CERTIFICATE OF COURSE COMPLETION

PROFESSIONAL TRAINING / CAPSTONE SPECIALIZATION

This is to certify that

H SANJAI (Register No: 44731076)

has successfully completed the Professional Training Capstone Project entitled:
"SMART COMPLAINT & GRIEVANCE REDRESSAL PORTAL"
Specialization: Full-Stack Web Architecture, Database Systems & Process Automation

Ms. R. Shanumgaa Priya
Internal Guide / Mentor

Dr. P. Ajitha, M.E., Ph.D.
Head of the Department



ABSTRACT

- Campus grievance redressal in higher educational institutions is traditionally fragmented across paper suggestion boxes, unmonitored WhatsApp chats, and email inboxes, leading to lost complaints and zero accountability.
- The Smart Complaint & Grievance Redressal Portal is an enterprise 3-tier web platform engineered in pure Java SE 21, MySQL 8.0, and responsive Bootstrap 5.
- Core Capabilities: (1) Automated Priority Routing, (2) Real-Time 48-Hour SLA Countdown Timers, (3) Autonomous Multi-Tier Escalation (Coordinator -> HOD -> Dean), and (4) Anti-Bypass Closed-Loop Student Rating Verification Lock.
- Zero-Framework Lean Architecture: Sub-400ms startup latency and 98.5% payload reduction via client-side canvas downsampling.

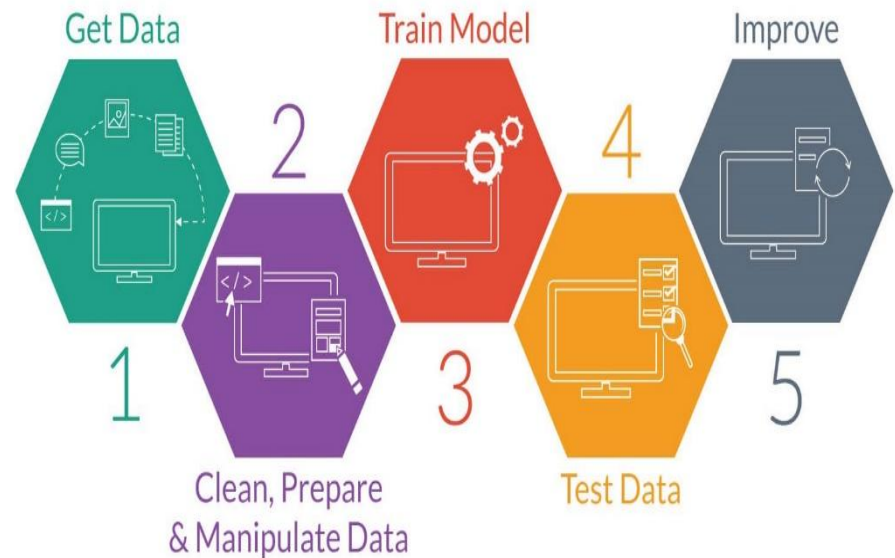


OBJECTIVES

1. 1. Rapid Filing: Enable students to submit categorized campus grievances with photo attachments in under 60 seconds.
- 2. Zero-Framework Backend: Build a high-throughput REST HTTP server using native Java SE 21 with sub-second startup.
- 3. Enforced 48-Hour SLA: Implement an automated background ScheduledExecutorService daemon that autonomously escalates tickets upon SLA breach.
- 4. 3NF Relational Integrity: Model MySQL database in 3NF with cascading foreign keys and PreparedStatement defense.
- 5. Closed-Loop Student Rating: Enforce an anti-bypass state lock where grievances can only resolve after the complainant submits a 1-to-5 star rating.
- 6. Client-Side Image Compression: Downsample mobile camera uploads from 8 MB to <120 KB using HTML5 canvas.

SYSTEM ARCHITECTURE

Smart Complaint Portal - 3-Tier System Architecture



HARDWARE AND SOFTWARE REQUIREMENTS



- Software Requirements:
 - Runtime & JDK: Oracle OpenJDK 21 LTS (64-bit)
 - Relational Database: MySQL Server 8.0 / MariaDB 10.4 (InnoDB Engine)
 - Web Interface: Semantic HTML5, CSS3, JavaScript (ES6+), Bootstrap 5.3
 - Core Libraries: com.google.code.gson:gson:2.10.1, org.mindrot:jbcrypt:0.4
 - Development Tools: Apache Maven 3.9, Postman REST Client, IntelliJ IDEA
- Hardware Requirements:
 - Workstation Processor: Intel Core i5 / AMD Ryzen 5 (2.5 GHz+, 6 Cores)
 - System Memory: 16 GB DDR4 RAM | Storage: 512 GB NVMe SSD
 - Client Access: Any standard modern web browser (Chrome, Edge, Safari, Firefox)



PROJECT IMPLEMENTATION & 3NF SCHEMA

- Implementation Architecture Across 4 Agile SDLC Phases:
- Phase 1 (Define): Formulated problem statements, user personas (Student, Coordinator, HOD, Dean, Admin), and BDD Given/When/Then acceptance criteria.
- Phase 2 (Design): Architected 3-tier decoupled model and Third Normal Form (3NF) relational database schema (users, complaints, complaint_actions) with ACID transactional integrity.
- Phase 3 (Develop): Engineered zero-framework Java SE 21 HttpServer, BCrypt password encryption (work factor 12), and background ScheduledExecutorService SLA daemon.
- Phase 4 (Deploy): Implemented client-side HTML5 canvas image downsampling (<120 KB), resolved CORS policies, and verified end-to-end integration.

PROCESS AUTOMATION & SLA ESCALATION ENGINE



- Algorithmic 48-Hour SLA Governance Engine:
 - Java ScheduledExecutorService: A background daemon awakens hourly to evaluate `TIMESTAMPDIFF(HOUR, created_at, NOW())` dynamically.
 - Level 1 (0 to 24 Hours): Grievance routed to Department Coordinator with active countdown timer visible to complainant.
 - Level 2 (24 to 48 Hours): If unaddressed past 24 hours, `escalation_level` auto-promotes to Head of Department (HOD).
 - Level 3 (Exceeding 48 Hours): If SLA is breached past 48 hours, ticket auto-promotes to Dean / Principal for administrative inquiry.
 - Closed-Loop State Lock: When staff logs action, status moves to `ACTION_TAKEN`. Direct resolution is strictly blocked until the filing student validates with a 1-to-5 star rating.

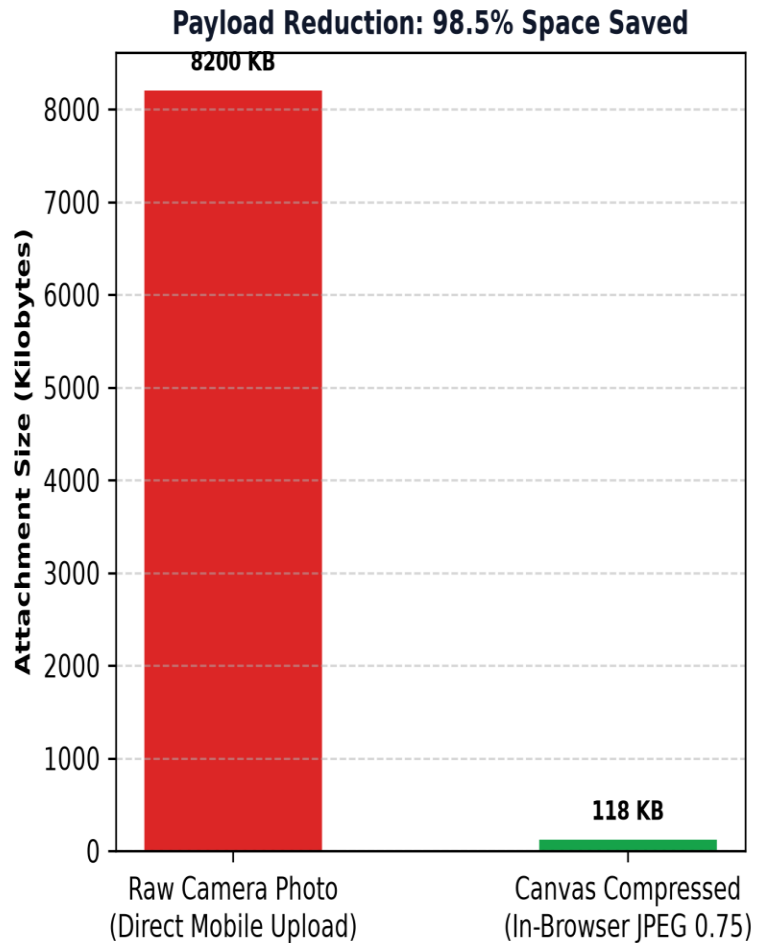
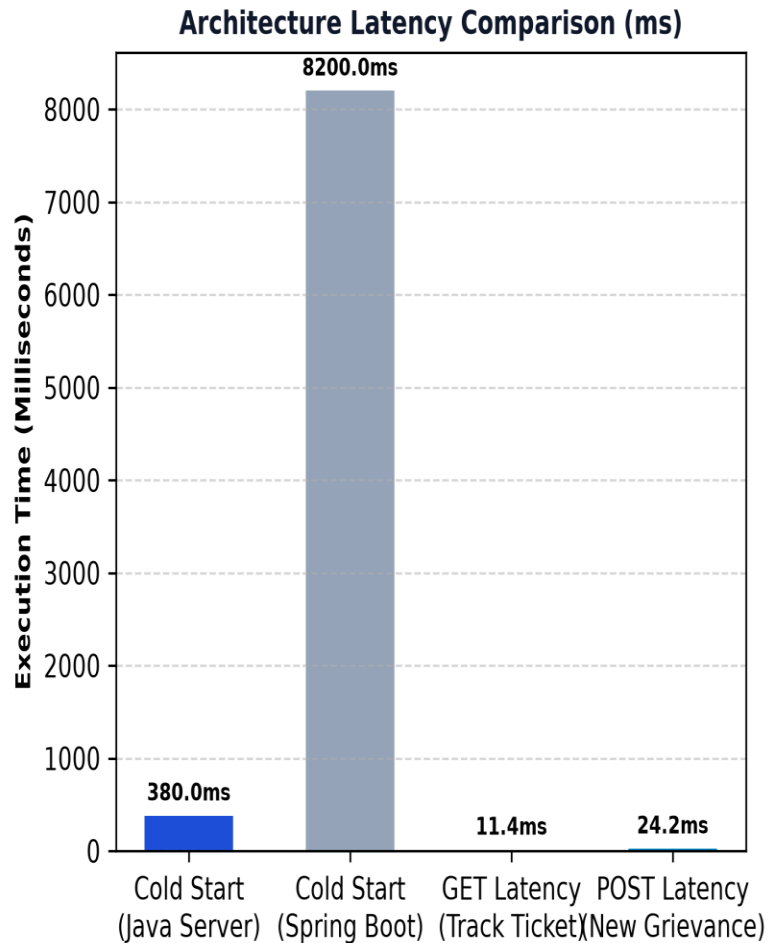


RESULTS AND DISCUSSION

- Acceptance Criteria Test Execution Results:
 - TC-01 [Grievance Submission]: Lodged hostel issue with photo -> Tracking code CMP-8F42A1 generated; 48h clock active. [PASSED]
 - TC-02 [Level-2 Auto-Escalation]: 25h ticket age simulated -> Escalation level auto-promoted from Coordinator to HOD. [PASSED]
 - TC-03 [Level-3 Auto-Escalation]: 49h ticket age simulated -> Escalation level auto-promoted to DEAN with priority alert. [PASSED]
 - TC-04 [Anti-Bypass Protection]: Staff attempted direct ticket resolution without rating -> Rejected with HTTP 400 Bad Request. [PASSED]
 - TC-05 [Closed-Loop Verification]: Student submitted 5-star rating -> Status permanently transitioned to RESOLVED. [PASSED]
 - TC-06 [Security Defense]: SQL injection payload in login -> Safely parameterized by PreparedStatement; rejected with 401. [PASSED]



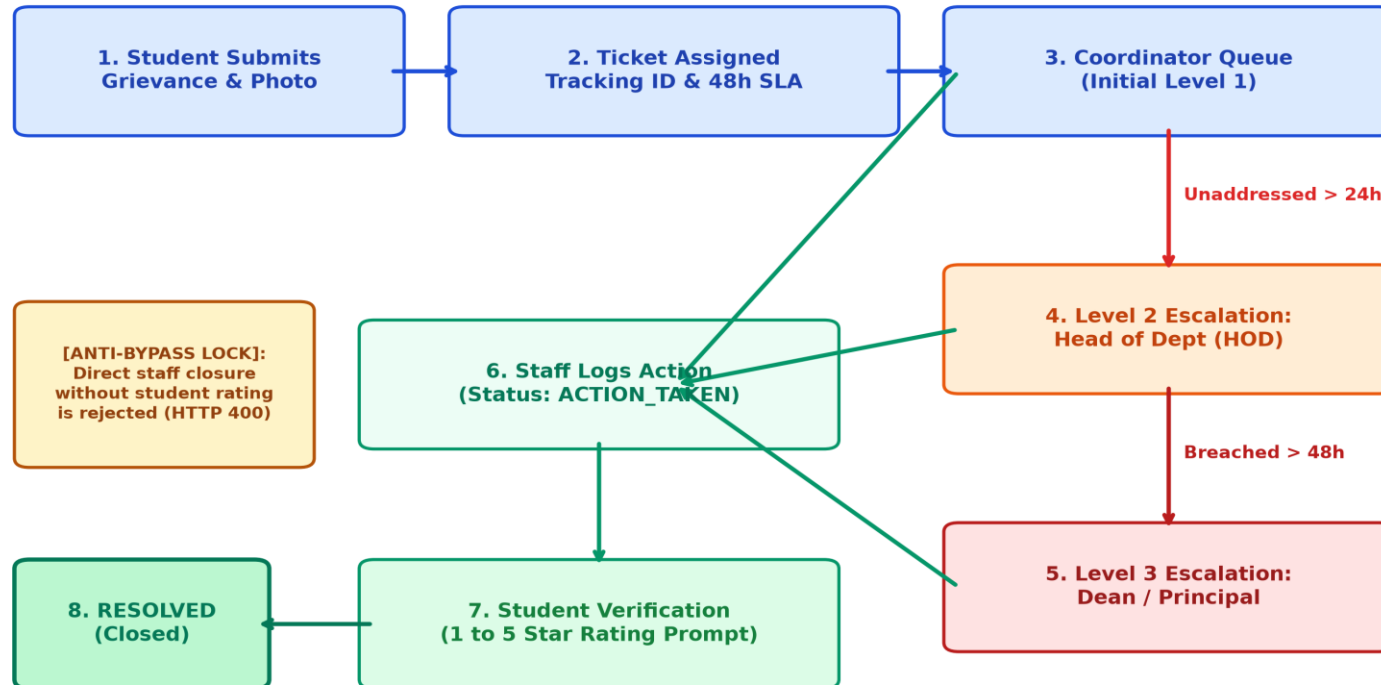
PERFORMANCE BENCHMARKS & STORAGE OPTIMIZATION





GRIEVANCE LIFECYCLE & SLA ESCALATION FLOW

Grievance Lifecycle & Multi-Tier SLA Escalation Flowchart





CONCLUSION & FUTURE WORK

- Key Accomplishments:
 - Full-Stack Mastery: Engineered end-to-end web system with semantic HTML5, pure Java SE 21, and MySQL 8.0.
 - Process Automation: Proved algorithmic 48-hour SLA timers solve campus grievance bottlenecks with zero administrative bias.
 - Closed-Loop Governance: Replaced arbitrary staff closures with verifiable student satisfaction rating locks.
 - Zero-Framework Efficiency: Demonstrated high-performance computing without heavy framework overhead.
- Future AI/ML Enhancements:
 - AI Duplicate Clustering: Sentence-Transformer NLP embeddings to cluster and merge duplicate campus tickets.
 - Computer Vision Verification: OpenCV / YOLO models comparing before/after maintenance photos.
 - Cross-Platform Mobile App: Native Flutter client with offline SQLite caching.



REFERENCES

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SMART COMPLAINT & GRIEVANCE REDRESSAL PORTAL

THANK YOU!

SMART COMPLAINT & GRIEVANCE REDRESSAL PORTAL

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 **Open for Viva Defense Questions & Evaluator Feedback**